REZA ASAD

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Research Interests

Designing and analyzing optimization algorithms for RL and ML; RLHF for LLMs

Education

Simon Fraser University **Jan 2020 – Oct 2026**

PhD in Computer Science, Optimization in RL and ML.
Thesis defended on October 20, 2026

Simon Fraser University **Sept 2018 – Dec 2019**

Master's in Computer Science, Visual Computing.

University of Toronto **Sept 2010 – Apr 2013**

Honours Bachelor of Science with High Distinction in Mathematics.

Recent Professional Experience

Applied Scientist II (L5) Intern at Amazon, Vancouver **June 2026 – Sept 2026**

- Developed Persona Variation Subspace (PVS), a low-rank framework for learning persona-conditioned preferences from sparse LLM annotations; reduced persona-level Brier score by approximately 16% relative to independent Bradley–Terry models in the sparsest setting.
- Established near-optimal prediction guarantees and conditions for exact recovery of aggregate preference cycles; work accepted at the EconML workshop at NeurIPS 2026, with a modified version under review at TMLR and a patent application filed through Amazon.

RL Scientist Intern at RBC Borealis, Vancouver **Jan 2026 – May 2026**

- Adapted Softmax Policy Mirror Ascent (SPMA), which I developed at Microsoft Research, for LLM post-training in RLHF and Nash learning from human feedback (NLHF), developing a tractable optimization formulation that avoids computing intractable normalization sums over responses.
- Implemented and evaluated the resulting algorithms, demonstrating improved training stability with less hyperparameter tuning compared with baseline methods.

RL Scientist Intern at Microsoft Research, Montreal **Mar 2024 – Jun 2024**

- Investigated functional Nesterov acceleration in RL, building on prior work that interprets parametric Nesterov acceleration as a combination of primal (gradient descent) and dual (mirror descent) progress.
- Framed mirror ascent as a general optimization framework for on-policy RL, recovering popular objectives and leading to the design of Softmax Policy Mirror Ascent (SPMA); accepted at AISTATS 2025.

Computer Vision Scientist Intern at Terramera, Vancouver **May 2019 – Aug 2019**

- Implemented a deep crowd-counting model to estimate corn crops from drone imagery, achieving 99.96% test accuracy; integrated into Terramera's product line.
- Developed a plant phytotoxicity segmentation model, achieving ~95% accuracy; used by Terramera's biologists.

Data Scientist at Tamr, San Francisco Bay Area **Oct 2016 – Jun 2018**

- Built a classifier for categorizing manufacturing parts (~17M weekly records) at General Electric, developing automated pipelines with Spark, Elasticsearch, and PostgreSQL, resulting in $80M in estimated savings over 3 years.
- Implemented an active learning framework to optimize label acquisition, maximizing the efficiency of domain experts' time.

Data Scientist at Albeado, San Francisco Bay Area **Nov 2015 - Oct 2016**

- Optimized a graph-based anomaly detection algorithm to identify provider fraud, directly securing Albeado's first customer (Flex LTD).
- Partnered with engineering to productionize the model, building a scalable automated pipeline for fraud detection.

Academic Research Experience

Doctoral Researcher, Simon Fraser University **Jan 2020 – Oct 2026**

- Developed a theoretically grounded policy gradient method, Softmax Policy Mirror Ascent (SPMA), bridging the gap between theory and practice in on-policy RL; published at AISTATS 2025.
- Identified a bottleneck in the Discrete Soft Actor-Critic (DSAC) objective and developed a general actor-critic framework that subsumes DSAC and extends our SPMA to the off-policy setting; published at a NeurIPS workshop and under review at TMLR.
- Conducted extensive empirical evaluations on Atari benchmarks and linear MDPs to validate a novel optimistic Actor-Critic framework against strong baselines; published at AISTATS 2026.
- Designed an optimization algorithm that reduces GPU idle time in large-scale model-parallel training while maintaining the convergence rate of gradient descent; published at a NeurIPS workshop.
- Introduced 3D subscene retrieval (3DSSR) and developed a self-supervised point-cloud encoder; improved geometric retrieval mAP by 9.11 percentage points over a prior self-supervised baseline on Matterport3D; presented as a spotlight at a CVPR 2023 workshop.

Master's Student, Simon Fraser University **Sept 2018 – Dec 2019**

- Modified the CycleGAN loss to translate between snowy and non-snowy satellite images while preserving cloud pixels, significantly improving cloud segmentation using the generated synthetic data.
- Won the Best Project Award at SFU's AI Showcase; an extension of the work was accepted to IEEE ICIP 2019.

Research Assistant, University of British Columbia **Sept 2013 – April 2015**

- Analyzed evolutionary processes through mathematical modeling, specifically extending the Moran model to complex population structures.
- Derived fixation probabilities to quantify the likelihood of mutant takeover.

Publications

1. "Dissecting Discrete Soft Actor-Critic: Limitations and Principled Alternatives", **R. Asad**, R. Babanezhad, S. Vaswani. *Under Review at Transactions on Machine Learning Research (TMLR), 2026.*

2. "Learning Persona-Conditioned Preferences and Recovering Cycles in Their Aggregate", **R. Asad**, M. Ghasemi, A. Banitalebi-Dehkordi. *Under Review at Transactions on Machine Learning Research (TMLR), 2026.*

3. "Sharing Statistical Strength Across Personas: Sample-Efficient Preference Learning and Aggregate Intransitivity", **R. Asad**, M. Ghasemi, A. Banitalebi-Dehkordi. *NeurIPS Workshop on Economics for Machine Learning (EconML), 2026.*

4. "Optimistic Actor-Critic with Parametric Policies for Linear Markov Decision Processes", M. Lin, **R. Asad**, K. Tan, H. Ishfaq, C. Szepesvari, S. Vaswani. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.*

5. "Revisiting Actor-Critic Methods in Discrete Action Off-Policy Reinforcement Learning", **R. Asad**, R. Babanezhad, S. Vaswani. *NeurIPS Workshop on Aligning Reinforcement Learning Experimentalists and Theorists (ARLET), 2025.*

6. "Fast Convergence of Softmax Policy Mirror Ascent", **R. Asad**, R. Babanezhad, I. Laradji, N. Le Roux, S. Vaswani. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2025*.

7. "Surrogate Minimization: An Optimization Algorithm for Training Large Neural Networks with Model Parallelism", **R. Asad**, R. Babanezhad, I. Laradji, N. Le Roux, S. Vaswani. *NeurIPS Workshop on Optimization for Machine Learning (OPTML), 2023.*

8. "3DSSR: 3D Subscene Retrieval", **R. Asad**, M. Savva. *CVPR Workshop (**Spotlight**) on Structural and Compositional Learning on 3D Data, 2023.*

9. "Steiner Symmetrization Along a Certain Equidistributed Sequence of Directions", **R. Asad**, A. Burchard. *arXiv, 2020.*

10. "CloudMaskGAN: A Content-Aware Unpaired Image-to-Image Translation Algorithm for Remote Sensing Imagery", S. Mohajerani, **R. Asad**, K. Abhishek, N. Sharma, A. van Duynhoven, P. Saeedi. *IEEE International Conference on Image Processing (ICIP), 2019.*

11. "Embedded Eigenvalues and the Nonlinear Schrödinger Equation", **R. Asad**, G. Simpson. *Journal of Mathematical Physics, 2011*.

Honors and Awards

- CMPT Graduate Fellowship Award, 2020-2024 ($34000).
- Dean's Graduate Scholarship, 2020-2024 ($5000).
- British Columbia Graduate Scholarship (BCGS), 2020-2024 ($15000).
- Best project award at SFU's AI showcase, Simon Fraser University, Fall 2018.
- Undergraduate Student Research Awards (USRA) through Natural Sciences and Engineering Research Council of Canada (NSERC), University of Toronto, Summer 2012 ($6000).
- Exceptional Achievement Award, University of Toronto, 2011-2012 ($1000).
- Later Life Learning OSOTF Award, University of Toronto, 2011-2012 ($600).
- Fairfax Financial Award, University of Toronto, 2008-2012 ($20000).

Technical Skills

- **Programming Languages:** Python (NumPy, PyTorch, OpenCV, pandas), C++ (OpenCV), R, MATLAB.
- **Tools:** PostgreSQL, Spark, Elasticsearch, MongoDB, Kafka, Amazon EC2.

Services

- Reviewer: TMLR (2026), NeurIPS (2025-2026), ICLR (2025-2026), OPTML Workshop at NeurIPS (2023-2026), ARLET Workshop at NeurIPS (2025), EconML Workshop at NeurIPS (2026).
- Teaching Assistant: Technical Writing and Research Communication (2021, 2023).
- Mentored Undergraduate Students**:**
    - Grayson Lee, SFU. *Next: Master's student at SFU*.
    - Amirhossein Etaati, SFU. *Next: Software Developer at RBC.*
- Maintained SFU's Graphics Union Vision (GrUVi) website from 2020-2023.